TECHNICAL REPORT

Supply Chain Intelligence & Feature Engineering for Predictive Logistics

SwiftLog Logistics Limited · Mission DS05 · Data Science Track

Item	Detail
Client	SwiftLog Logistics Limited
Prepared by	Sharin Mideva, Junior Data Scientist
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1. Executive Summary

SwiftLog Logistics Limited is expanding quickly and is seeing delivery delays, inventory imbalances and operational bottlenecks. Before investing in predictive analytics and AI, management asked for a clear picture of what its operational data says. This report integrates the company's 11 data tables (5,000 orders, 1,000 customers, 15 warehouses, 250 products, 15,000 order lines, 5,000 shipments and 500 returns), cleans and validates them, explores them statistically, engineers features for future machine learning and segments customers and warehouses.

Key findings

- **Structurally sound, content-incomplete data.** All 11 tables load cleanly: 0 missing values, unique primary keys and 0 orphan foreign keys. But suppliers link to no other table, there is no warehouse capacity, and only the order date exists (no ship, delivery or promised dates).
- **Revenue is large but leaky.** Net revenue (excluding cancelled orders) is 14.44M against 21.67M gross. Cancelled orders hold 7.23M, about a third of gross, and only 33% of orders are Delivered (34% Pending, 32.5% Cancelled).
- **Delivery delay is systemic.** 63.7% of shipments are 3+ days late, spread almost evenly across 0 to 7 days. No warehouse ($p = 0.328$) or customer city ($p = 0.099$) differs significantly, and delay is unrelated to distance, workload or returns.
- **One route stands out.** Kano warehouses serving Lagos customers are delayed only 53.4% of the time (249 orders, Bonferroni-adjusted $p = 0.008$), about 10 points better than average. It is the best lead for improvement.
- **Customer value is concentrated in one segment.** S1 (Loyal high-value) is 35% of customers but 59% of net revenue; S3 (Low-frequency, low-value) is 43% of customers, 23% of revenue and has gone 61 days since its last order.
- **Returns are a constant ~9.5% background rate.** Late Delivery (37%), Wrong Item (33%) and Damaged (30%) are the stated reasons, but 36% of "Late Delivery" returns came from shipments classed On Time, so the reason field is unreliable.
- **Inventory does not reconcile with activity.** Stock held is about 28 times units sold, 34.4% of shipped warehouse-product pairs have no inventory record, and 10.8% of stocked pairs hold fewer than 100 units.

Priority recommendations

Area	Recommendation
Delivery	Replicate the Kano-to-Lagos route process; launch a company-wide delay-reduction programme.
Cancellations	Capture cancellation reasons and age pending orders; target the 7.2M cancelled value.
Data capture	Record promised and actual delivery dates, supplier links, warehouse capacity and valid return reasons.
Inventory	Reconcile stock records with shipments, set reorder levels and rebalance stock between warehouses.
Customers	Protect S1 with priority service; run win-back and basket-building offers for S3.
Returns	Add picking verification and packaging checks; follow up with return-prone S2 customers.

How far to trust the results

Delay days, quantity and unit price are uniformly distributed and most categories are almost equal in size, which suggests the data may be simulated. Five of six hypothesis tests found no significant difference, and this report treats those as genuine null results and not as gaps to explain away. Engineered customer and warehouse features were computed over all orders and must be rebuilt using only earlier data before any predictive model is trained. Thirteen recommendations are given in section 9, each tied to evidence.

2. Data Understanding

Twelve CSV files were supplied: eleven data tables and a data dictionary. The dictionary lists the files only and does not define fields, so meanings were inferred from column names and values and documented in Appendix A. The table below summarises each dataset, its keys and how it connects to the others.

Table	Rows	Cols	Primary key	Foreign keys	Notes
customers	1,000	3	CustomerID	none	Name and city (Abuja, Ibadan, Kano, Lagos)
orders	5,000	4	OrderID	CustomerID	Order date and status; 1 Jan to 8 Sep 2026
order_items	15,000	4	OrderItemID	OrderID, ProductID	Quantity per line; no price column
products	250	3	ProductID	none	Category (Fashion, Food, Electronics) and unit price
shipments	5,000	4	ShipmentID	OrderID, WarehouseID	One shipment per order; DelayDays 0 to 7
delivery_performance	5,000	3	ShipmentID	ShipmentID	On Time / Delayed status; DelayDays matches shipments
warehouses	15	2	WarehouseID	none	Lagos 8, Abuja 4, Kano 3
inventory	4,000	3	WarehouseID + ProductID (composite)	WarehouseID, ProductID	Repeats are stock movements: 4,000 rows become 2,455 pairs
employees	200	2	EmployeeID	WarehouseID	9 to 17 staff per warehouse
returns	500	3	ReturnID	OrderID	Reason only; no product, no date
suppliers	150	2	SupplierID	none	Isolated: no other table references it

Key validation findings:

All 10 primary keys are unique and non-null. All 10 foreign key relationships have zero orphan records, so referential integrity holds. `inventory` has a composite key (WarehouseID + ProductID), but repeated combinations are stock movements, aggregated by summing Stock (4,000 rows to 2,455). `suppliers` has no link to any other table and is analysed standalone.

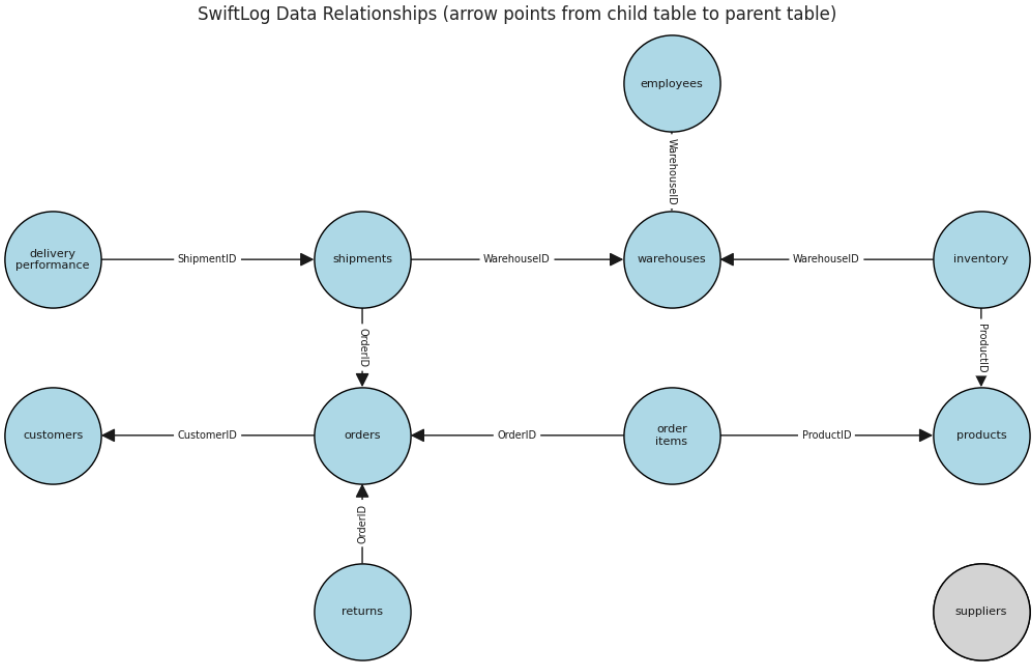


Chart 00: Data relationship diagram (arrows point from child table to parent table; suppliers, in grey, has no links).

Data quality summary

Check	Result
Missing values	0 in every table
Duplicate rows	0 in every table except inventory (3 identical rows, treated as stock movements)
Primary keys	All 10 unique and non-null
Foreign keys	10 relationships checked, 0 orphan records
Delay columns	DelayDays in shipments and delivery_performance agree on all 5,000 rows
Text categories	Already consistent: standardisation changed 0 values
Date range	OrderDate parses on all rows: 2026-01-01 to 2026-09-08 (September is partial)

3. Data Cleaning and Preparation

Because the data had no missing values, label inconsistencies or key problems, most cleaning work was validation. The material changes were converting dates, aggregating inventory movements, defining the delay flag and revenue, and building analysis-ready datasets. Every step is logged in `cleaning_log.csv` in the notebook outputs.

Data Quality Findings and Cleaning Decisions

The checks run so far show the data is structurally sound but contains several logical inconsistencies. Each finding below is paired with the decision taken.

Check	Result	Decision
Text standardisation	0 values changed in every categorical column	Labels were already consistent. Standardisation was run as a validation step.
Delay flag	`DelayDays >= 3` matches the supplied `Status` on all 5,000 shipments	The rule "Delayed = 3 or more days late" is an inference confirmed by the data. It is used as the delay definition throughout.
Repeated returns	473 orders have returns: 446 have one, 27 have two. 5 rows repeat the same order and reason	All rows kept. Each has a unique `ReturnID` and there is no date to prove a repeat is a duplicate.
Orders without items	236 of 5,000 orders have no rows in `order_items`	Revenue set to 0 and a `HasItems` flag added. Average order value is calculated only over orders with items.
Outliers (IQR rule)	`Quantity`, `UnitPrice`, `DelayDays`: 0. `LineRevenue`: 119. `Stock`: 87	Kept. `LineRevenue` is quantity x price, so large lines are natural, and `Stock` is a sum of movements, so pairs with more movements sum higher. Removing them would delete real business activity.

Conclusion:

Because there were no missing values, no label inconsistencies and no key problems, most cleaning work was validation. The material changes were converting `OrderDate` to datetime, aggregating inventory stock movements, and building calculated fields. Results of checks that found no problems are reported as such and are not treated as errors to fix.

Revenue Definition

The dataset has no revenue or amount column. Revenue is therefore calculated:

- **Line revenue** = `Quantity` x `UnitPrice`, where the unit price comes from the products table.
- **Order revenue** = the sum of line revenue across all items in an order.

Orders in this dataset have one of three statuses: Delivered, Pending, or Cancelled. Counting all of them as sales would overstate performance, so three measures are reported:

Measure	Definition	Amount
Gross revenue	All orders, regardless of status	21,668,066
Net revenue (headline)	All orders excluding Cancelled	14,439,218
Delivered revenue	Delivered orders only (conservative view)	7,107,471

Revenue by order status

Status	Orders with items	Revenue	Average order value
Cancelled	1,542	7,228,848	4,688
Delivered	1,578	7,107,471	4,504
Pending	1,644	7,331,748	4,460

What this tells us

1. **Net revenue is the headline figure** because cancelled orders were never completed sales. Delivered revenue is the most conservative figure, since Pending orders have not yet been fulfilled.

2. **Cancelled orders carry about a third of gross revenue** (7.2M of 21.7M, roughly 33%). This is a significant business finding on its own: a large share of order value is lost to cancellation.
3. **Order values are similar across statuses** (about 4,460 to 4,690 on average), so cancellation does not appear to be driven by order size.
4. **236 orders have no items** and contribute zero revenue. They are kept in order counts but excluded from average order value.

Assumptions

- The dataset does not state a currency. Figures are assumed to be in Naira, since SwiftLog is a Nigerian company.
- Order status and delivery status are independent in this data (for example, Cancelled orders have shipment records). Revenue follows order status, while delivery analysis uses shipment records regardless of order status.

Dataset Integration

The 11 tables sit at different levels of detail, so they were merged into two analytical datasets and not one. Merging everything into a single table would repeat order-level values (such as shipment delay) once for every item in the order and inflate counts and totals.

Dataset	One row per	Rows	Columns	Used for
`order_master`	Order	5,000	25	Delivery, customer, warehouse and returns analysis
`item_master`	Order line	15,000	19	Product and category analysis

How `order_master` was built: orders joined to customers, shipments, delivery performance, and warehouses, then to order-item and return summaries. The order items and returns were **aggregated to order level before joining**, so each order stays on a single row. Every join used pandas' `validate` option to confirm the relationship type (for example, one order to one shipment), so a wrong assumption would have raised an error.

Integrity checks (all passed):

- `order_master` has exactly 5,000 rows and `OrderID` is unique.
- `item_master` has exactly 15,000 rows.
- Order-level revenue reconciles exactly to line-level revenue (gross 21,668,066).
- No order is missing a customer, warehouse or delivery record.

Not integrated: `suppliers` has no key linking it to any other table, so it remains a standalone reference table. `inventory` and `employees` are analysed at warehouse level rather than order level.

Derived Columns and Encoding

Columns created

Column	How it was built	Purpose
`Route`	Warehouse location joined to customer city (e.g. "Lagos to Ibadan")	Compare delivery performance by shipping route
`SameCityDelivery`	1 if warehouse location equals customer city, otherwise 0	Test whether distance affects delays
`CustomerNumber`	Numeric part extracted from `CustomerName`	Split-column example; low analytical value
`HasReturn` / `ReturnCount`	Returns counted per order	Order-level return analysis

Finding: shipping distance does not explain delays.

Delivery type	Share of orders	Delay rate
Different city	75.1%	63.7%
Same city	24.9%	63.9%

The delay rates are effectively identical, so delivering from a warehouse in the customer's own city made no difference. The cause of delays therefore lies elsewhere, such as warehouse operations, and this guides where later analysis should look. Ibadan customers are always served from another city because Ibadan has no warehouse.

Encoding

Categorical variables were one-hot encoded into a separate dataset, `order_master_encoded` (5,000 rows, 36 columns), so `order_master` stays readable for analysis and charts. Encoded columns: order status, customer city, warehouse location and order quarter.

- **Identifier columns** (`OrderID`, `CustomerID`, `ShipmentID`) were not encoded, because they identify records and are not categories.
- `DeliveryStatus` **was dropped from the encoded set.** It carries the same information as `IsDelayed`, and keeping both would be redundant and would leak the answer in a future delay-prediction model.

Automated Data Profiling

An automated profiling report was generated with `ydata-profiling` on the final `order_master` dataset (identifier columns excluded) and saved as `Reports/Data_Profiling_Report.html`. It confirms the earlier manual checks: no missing values, no duplicate rows, and the distributions and correlations of every variable in one place.

4. Exploratory Data Analysis

The analysis uses 28 charts. Each carries an interpretation and a business implication. Net revenue (excluding cancelled orders) is the headline revenue measure, and delayed means three or more days late.

Univariate analysis

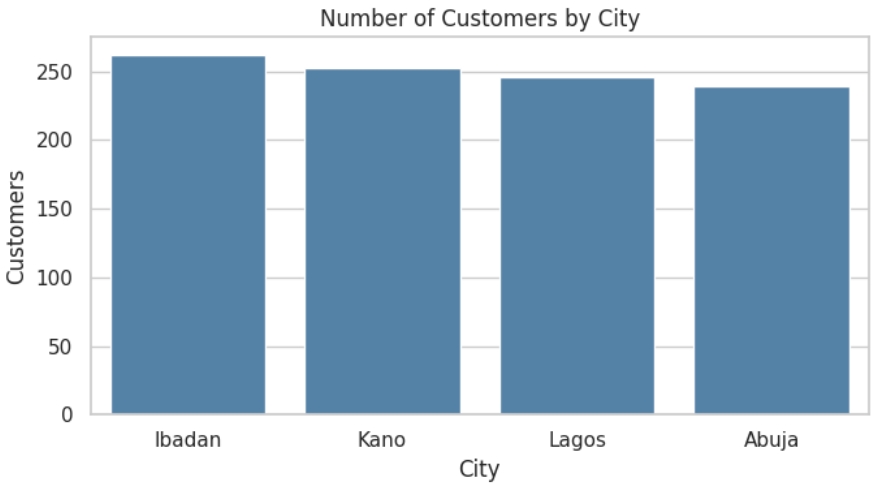


Chart 01: Customers by city

Interpretation: The 1,000 customers are spread almost evenly across four cities: Ibadan 262 (26.2%), Kano 253, Lagos 246 and Abuja 239 (23.9%). The gap between the largest and smallest city is only 23 customers.

Business implication: Demand is not concentrated in one market, so no city can be ignored. Ibadan has the largest customer base but no warehouse, so its customers are always served from another city.



Chart 02: Orders per customer

Interpretation: 991 of 1,000 customers placed at least one order, averaging 5.05 orders each (median 5, maximum 15). Only 40 customers (4%) ordered once, and 9 never ordered.

Business implication: Repeat buying is strong: about 96% of ordering customers bought more than once, a healthy retention signal. The 9 inactive customers and 40 one-time buyers are a small group for re-engagement, and the heaviest buyers (up to 15 orders) are candidates for the high-value segment in the segmentation section.

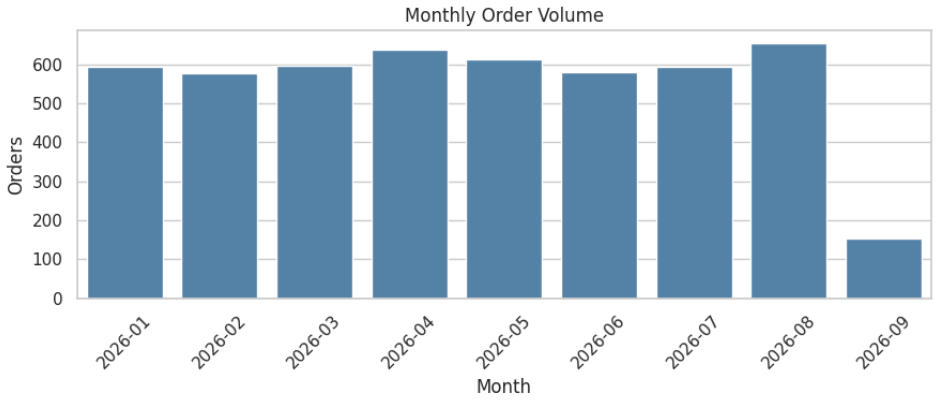


Chart 03: Monthly order volume

Interpretation: Full months (January to August) ranged from 577 to 654 orders, averaging about 606. August was the peak (654, roughly 8% above average), followed by April (638). February was the lowest (577). September shows 154 orders in only 8 days, so its low bar reflects a partial month and not a decline.

Business implication: Demand is steady with only small peaks in April and August, so capacity can be planned around roughly constant monthly volume. With only 8 months of data and no earlier year, a true seasonal pattern cannot be confirmed.

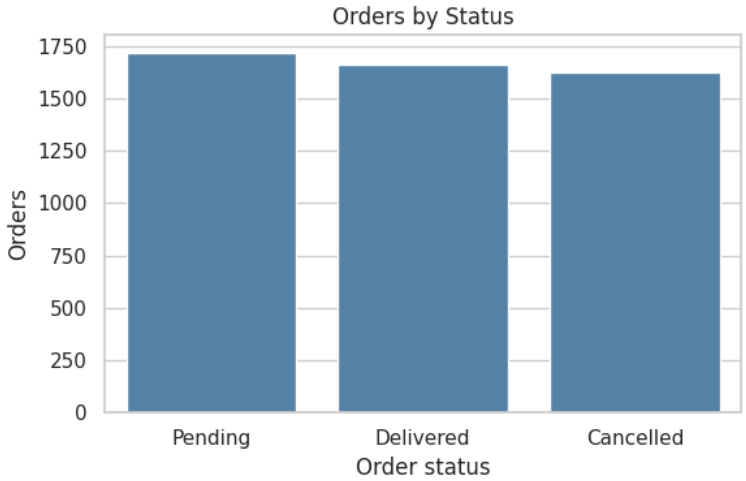


Chart 04: Orders by status

Interpretation: Orders divide almost exactly into thirds: Pending 34.3%, Delivered 33.2% and Cancelled 32.5%.

Business implication: Only a third of orders are delivered and nearly a third are cancelled, which is a serious fulfilment problem if the statuses are accurate. The 34% still Pending is also questionable, since the data spans 8 months and older orders should have been completed. Cancelled orders hold about a third of gross revenue (see section 2.2). How long orders stay pending should be investigated.

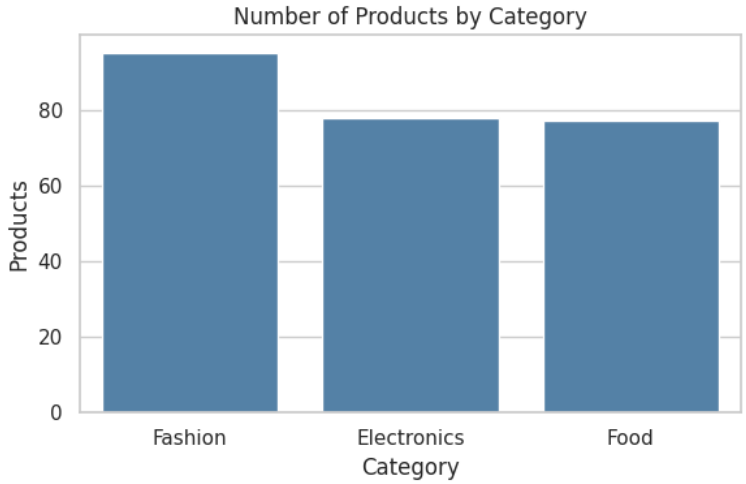


Chart 05: Products by category

Interpretation: Of 250 products, Fashion has 95 (38%), Electronics 78 (31.2%) and Food 77 (30.8%).

Business implication: The catalogue leans towards Fashion, but the number of products might not equal sales. Further bivariate analysis to be conducted.

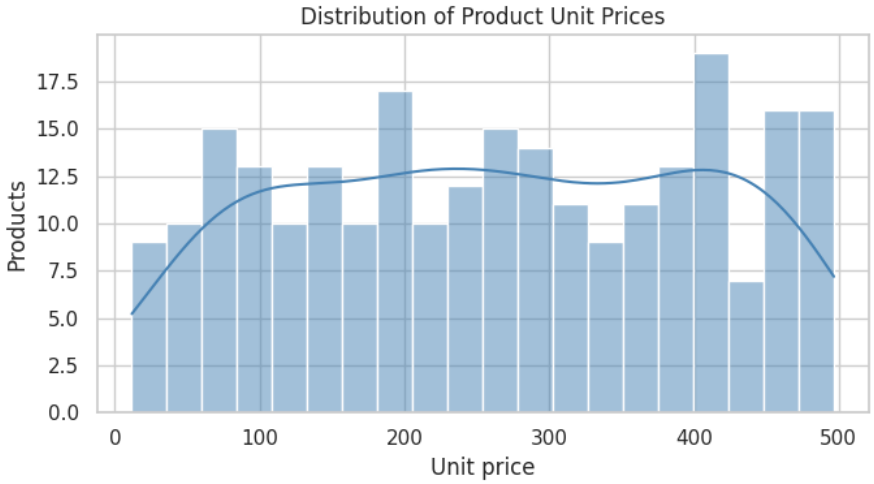


Chart 06: Unit price distribution

Interpretation: Unit prices range from 11.78 to 496.61, with a mean of 264.2 and a median of 263.6. The near-identical mean and median show a symmetric distribution with no extreme high-priced items.

Business implication: The catalogue covers the full price range and not a niche, and prices are not a source of outliers. Revenue will therefore be driven more by quantities ordered and product mix than by a few premium items.

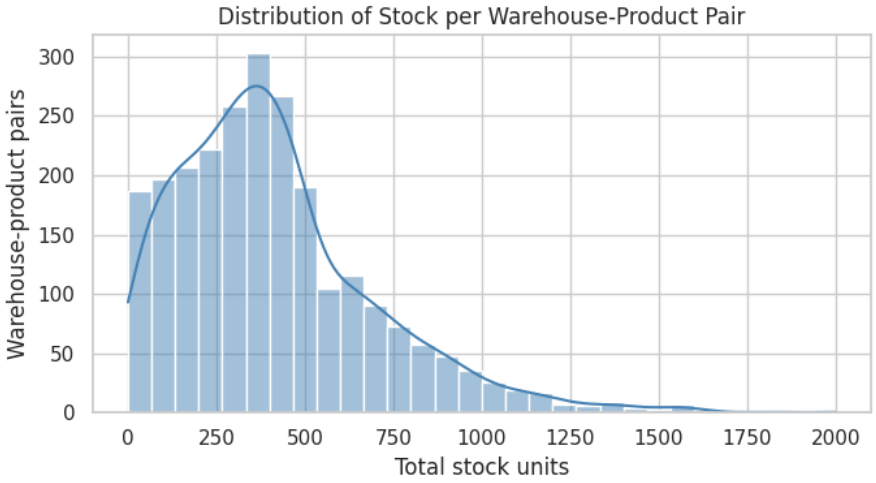


Chart 07: Stock per warehouse-product pair

Interpretation: Stock per warehouse-product pair averages 410.5 against a median of 364, a right-skewed distribution in which some pairs hold much more stock than the rest. 264 of 2,455 pairs (10.8%) hold fewer than 100 units.

Not every warehouse-product combination has an inventory record: 2,455 of the 3,750 possible pairs (15 warehouses x 250 products) do, so 1,295 pairs (34.5%) have none. A cross-check against shipments shows that 34.4% of the warehouse-product pairs that actually shipped also had no inventory record, almost exactly the share expected if shipments ignored inventory.

Business implication: About one in ten stocked pairs is at risk of running out (using an assumed threshold of 100 units, since no reorder level was provided). More importantly, inventory records do not line up with shipping activity: a third of shipped items have no record in the dispatching warehouse. Either the inventory data is incomplete, or orders are being fulfilled from warehouses that do not hold the product. Both possibilities weaken stock visibility and should be investigated before inventory figures are used for replenishment decisions.

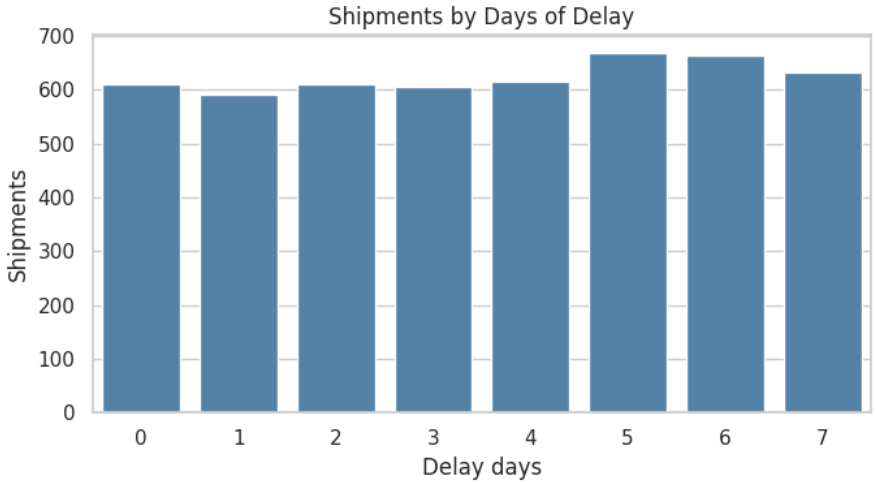


Chart 08: Shipments by days of delay

Interpretation: Delay is spread almost evenly across 0 to 7 days, with 592 to 668 shipments at each value (about 12.5% each). The mean delay is 3.57 days and 63.7% of shipments are delayed (3+ days), which is what an even spread would produce (5 of 8 values = 62.5%). Days 5 (668) and 6 (664) are slightly the most common.

Business implication: Delays are not occasional exceptions: nearly two in three shipments arrive 3 or more days late, with no typical delay length. That points to a systemic process problem and not isolated incidents. Even shipments classed as On Time can be up to 2 days late, so the on-time definition is generous. Delivery delay is the first KPI needing management attention.



Chart 09: Returns by reason

Interpretation: Of 500 returns, Late Delivery accounts for 185 (37.0%), Wrong Item 163 (32.6%) and Damaged 152 (30.4%).

Business implication: Late delivery is the largest single return reason, linking delivery performance directly to returns and customer dissatisfaction, and it is addressable through delivery improvements. Wrong Item (picking accuracy) and Damaged (handling and packaging) together make up 63% of returns and point to warehouse quality control. The bivariate analysis will test whether delayed orders actually have more returns.

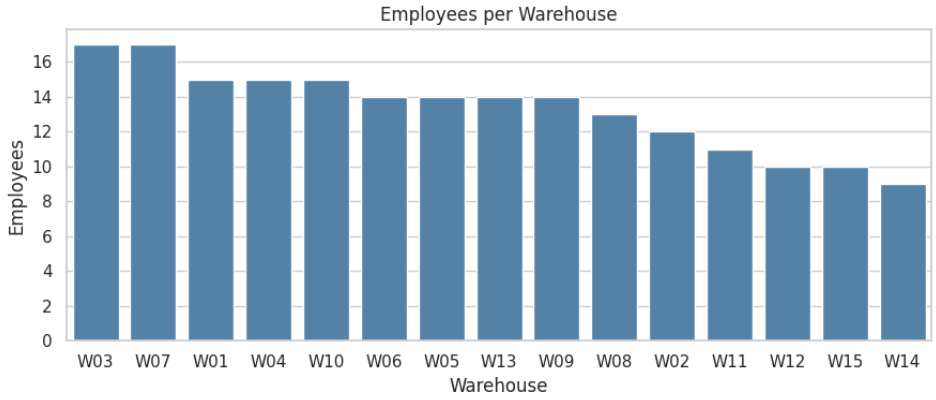


Chart 10: Employees per warehouse

Interpretation: The 200 employees are spread unevenly across 15 warehouses, from 9 to 17 per warehouse (average 13.3). W03 and W07 have the most (17 each), while W14 has only 9 and W12 and W15 have 10 each.

Business implication: The largest team is nearly double the smallest. Whether this is justified depends on workload, which is examined with chart 11.



Chart 11: Shipments per warehouse

Interpretation: Shipments per warehouse average 333. W07 handles the most (371, about 11% above average) and W13 the fewest (279, about 16% below). Most warehouses fall between 318 and 354.

Combining charts 10 and 11 gives shipments per employee (company average: 25):

Warehouse	Shipments	Employees	Shipments per employee
W14	340	9	37.8
W15	352	10	35.2
W12	337	10	33.7
W13	279	14	19.9
W03	344	17	20.2

Business implication: Shipment volume is fairly balanced, but staffing is not. W14, W15 and W12 carry about 35% to 50% more shipments per employee than average, while W13 and W03 carry about 20% less. This is an early sign of workload imbalance and a candidate explanation for delays, to be tested in the bivariate analysis. Staff could be rebalanced from lightly loaded warehouses to the heavily loaded ones.

Suppliers: The suppliers table holds 150 records with only an ID and a name. With no performance variable and no link to any other table, no univariate analysis is possible.

Summary of Univariate Findings

1. **Demand is steady and evenly spread** across cities and months, with only small peaks in April and August.
2. **Customers are loyal:** about 96% of ordering customers bought more than once.
3. **Fulfilment is weak:** only 33% of orders are delivered, 32.5% are cancelled and 34% remain pending.
4. **Delays are systemic:** 63.7% of shipments are 3+ days late, spread evenly across 0 to 7 days.
5. **Late delivery is the top return reason** (37%).
6. **Staffing is uneven relative to workload:** W14, W15 and W12 handle the most shipments per employee.
7. **About 35% of warehouse-product combinations have no inventory record,** and about 11% of stocked pairs are below 100 units.

Caution: Many distributions here are unusually even, which suggests the data may be simulated. Small differences between categories should not be over-interpreted, and several later tests may show no significant difference.

Bivariate Analysis



Chart 12: Orders placed vs net revenue per customer

Interpretation: Customers who place more orders generate more net revenue (correlation 0.67), but the link is only moderate because order values and cancellations vary. C0773 placed 12 orders and earned 54,174,

while C0223 placed 8 orders and earned 58,660, the highest in the business. Revenue is not concentrated: the top 10% of customers (100 customers) generate 23.9% of net revenue, and the largest single customer accounts for only about 0.4%.

Business implication: SwiftLog does not depend on a few large accounts, which lowers risk, but there is also no small group of "VIP" customers to protect. Customer value depends on both how often customers buy and how large each order is. The top five customers (C0223, C0773, C0807, C0599, C0880) are natural candidates for a priority-account programme.

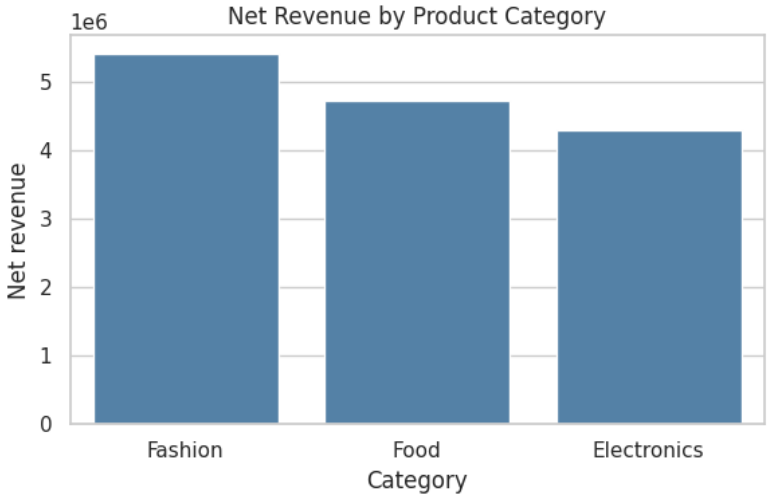


Chart 13: Net revenue by product category

Interpretation: Fashion earns the most net revenue (5.41M, 37.5%), followed by Food (4.74M, 32.8%) and Electronics (4.29M, 29.7%). Comparing each category's revenue share with its share of the catalogue: Fashion holds 38.0% of products and 37.5% of revenue (proportional), Food holds 30.8% of products but 32.8% of revenue, and Electronics holds 31.2% of products but only 29.7% of revenue.

Business implication: Fashion leads mainly because it has the largest range, not because each product performs better. Food earns more than its size suggests and Electronics slightly less, but the gaps are modest.

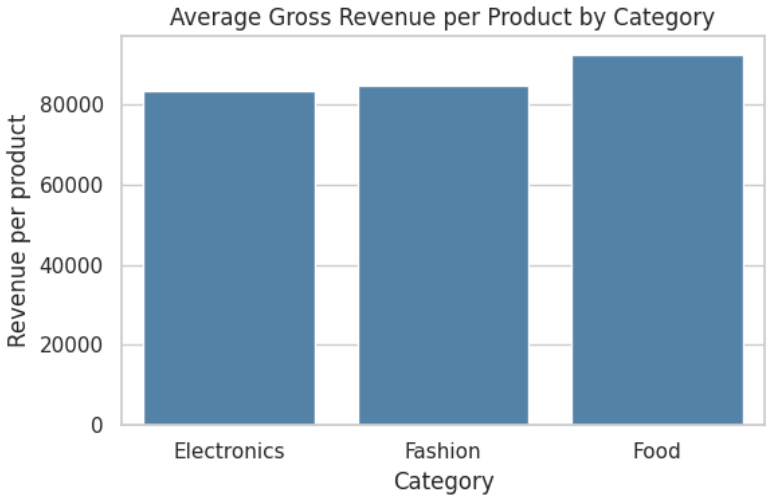


Chart 14: Average gross revenue per product by category

This chart uses gross revenue (all orders, including Cancelled) so that products are compared on everything customers ordered.

Interpretation: Food earns the most revenue per product (92,382), ahead of Fashion (84,838) and Electronics (83,270), a spread of about 11%. Fashion sells the most units (31,715 vs about 25,000 for each of the others), but Food earns more per unit sold (about 278 vs 254 for Fashion and 258 for Electronics), so Food's advantage comes from the value of what it sells, not from volume.

Business implication: Food is the most productive category per product, so extending the Food range or promoting it is worth testing. Fashion's total lead comes from having 95 products, not from stronger products.

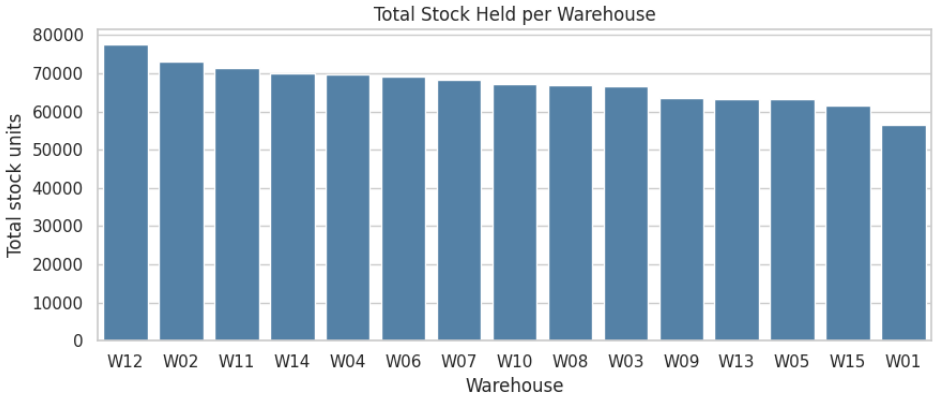


Chart 15: Total stock held per warehouse

Interpretation: Stock held ranges from 56,539 units at W01 to 77,517 at W12 (average about 67,200), so W12 holds about 15% above average and W01 about 16% below. Products stocked per warehouse range from 154 (W01) to 174 (W12) out of 250. W12 and W01 shipped almost the same number of orders (337 and 332), yet W12 holds about 230 units per shipment and W01 about 170.

Business implication: Stock is not distributed in line with demand. W12 ties up more inventory than its volume needs, while W01 is the thinnest and most exposed to stock-outs. Rebalancing stock towards warehouses with low stock per shipment could cut holding costs and shortages. This is indicative only, because inventory records are incomplete.

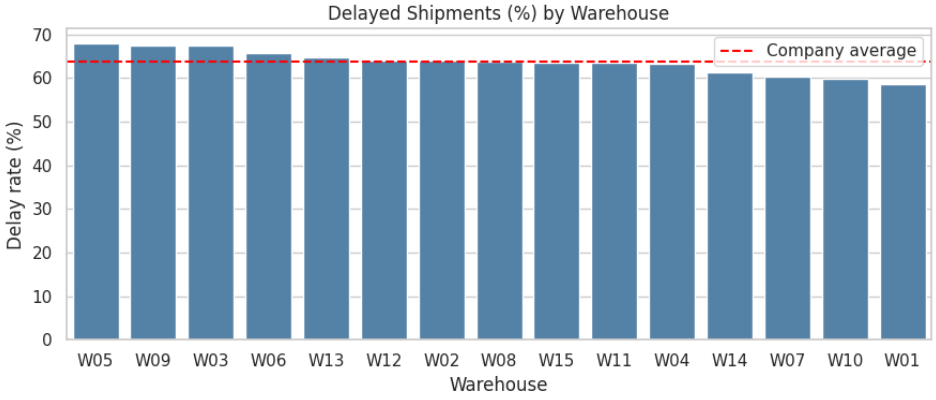


Chart 16: Delayed shipments (%) by warehouse

Interpretation: Every warehouse delays more than half its shipments, from 58.7% at W01 to 68.0% at W05 (company average 63.7%). W05 also has the longest average delay (3.79 days), followed by W09 and W03 (67.4% to 67.5%). The lowest rates are at W01, W10 (59.9%) and W07 (60.4%), and W07 achieves this while handling the most shipments (371). With about 330 shipments per warehouse, differences of around 5 points can easily arise by chance, and the best-to-worst spread is roughly what chance alone would produce for 15 warehouses.

Business implication: Delay is a company-wide problem, not the fault of a few poor warehouses. W01, W10 and W07 are the most efficient on this measure and their processes are worth studying, but no warehouse is performing well. Staffing does not explain the pattern either: W03 and W07 both have the largest teams (17), yet W03 is among the worst and W07 among the best.

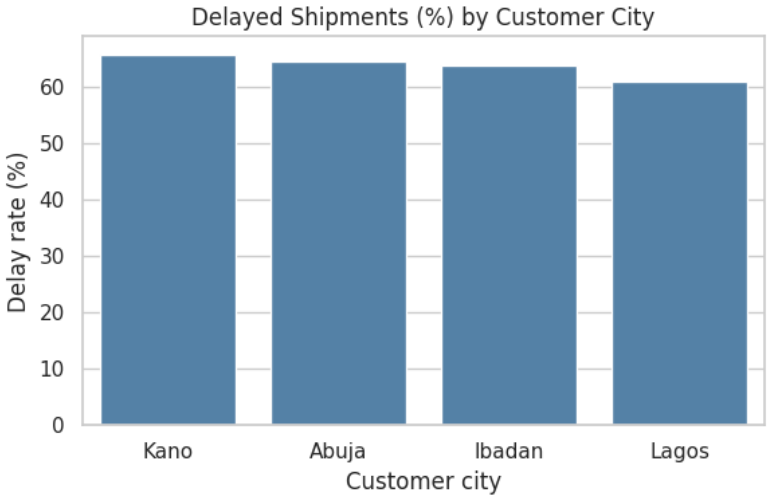


Chart 17: Delayed shipments (%) by customer city

Interpretation: Kano customers see the highest delay rate (65.7%), then Abuja (64.4%) and Ibadan (63.8%), while Lagos is lowest (61.0%). The Lagos-to-Kano gap is 4.7 points, on roughly 1,250 orders per city. Ibadan, which has no warehouse of its own, performs in line with the other cities.

Business implication: Delays are high everywhere and differences between cities are small. Since Ibadan customers are not worse off for lacking a local warehouse, opening a warehouse there is not justified by delivery performance alone.

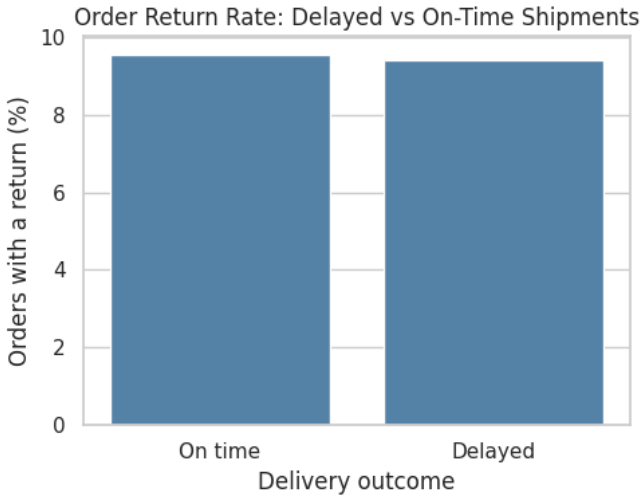


Chart 18: Order return rate, delayed vs on-time shipments

Interpretation: Orders with delayed shipments are returned at 9.42%, and orders delivered on time at 9.54%. The two are essentially identical.

Business implication: Delivery delay does not make a return more likely in this data. Overall about 9.5% of orders have a return regardless of how delivery went, so reducing delays will not by itself reduce returns.

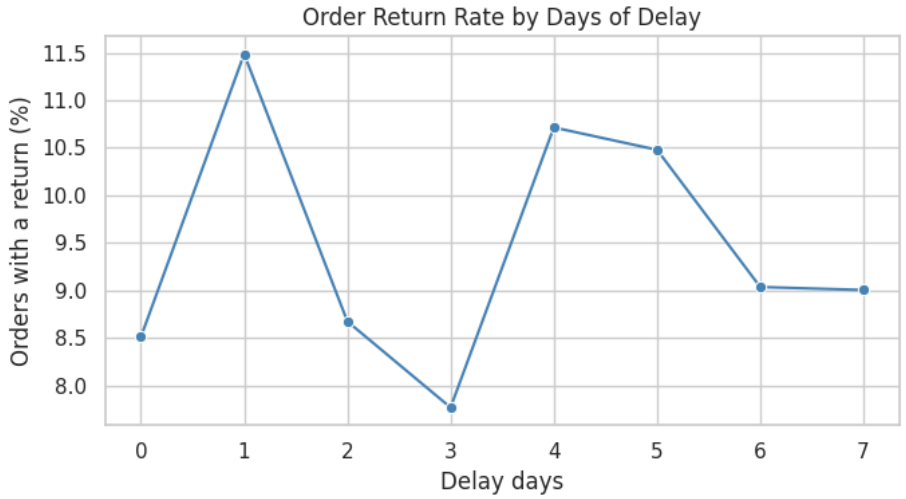


Chart 19: Order return rate by days of delay

Interpretation: The return rate moves between 7.8% (3 days) and 11.5% (1 day) with no trend, and the correlation between delay days and returns is effectively zero. Each delay-day group has roughly 600 orders, so swings of about 2 points are within normal chance variation.

Business implication: A longer delay does not lead to more returns, so there is no delay threshold at which returns start to rise. Returns need to be explained by other factors, such as picking errors and damage. A check on whether "Late Delivery" returns actually come from late orders follows below.

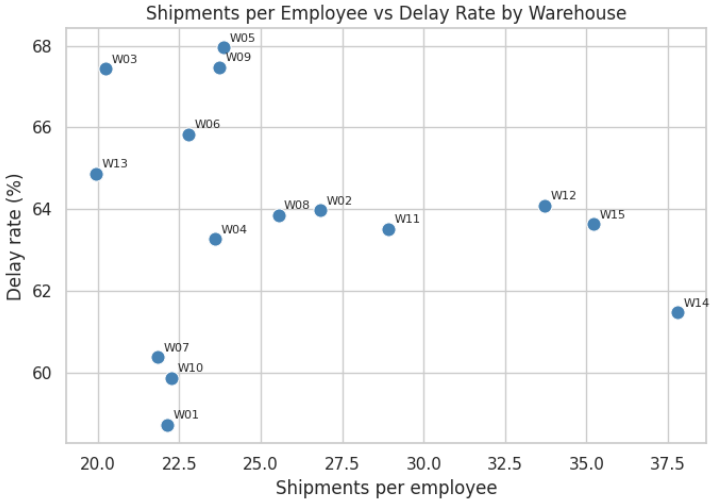


Chart 20: Shipments per employee vs delay rate by warehouse

Interpretation: Across the 15 warehouses the correlation between workload (shipments per employee) and delay rate is -0.13, which is weak and slightly in the opposite direction to expectation. The warehouses carrying the heaviest load, W14 (37.8 shipments per employee), W15 (35.2) and W12 (33.7), delay 61.5%, 63.6% and 64.1% of shipments, at or below the company average.

Business implication: The workload-imbalance explanation proposed after chart 11 is not supported. Delays are not caused by understaffed warehouses, so adding staff is unlikely to fix them on its own. Causes must lie in processes the data does not capture, such as dispatch handling, carrier performance or order processing time.

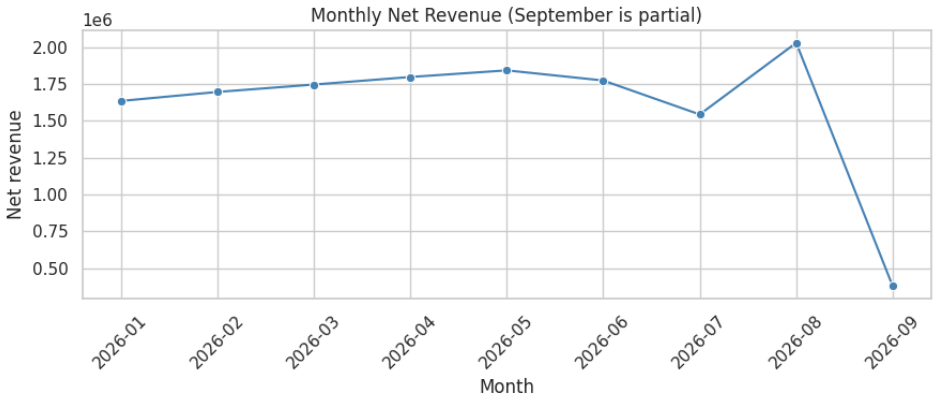


Chart 21: Monthly net revenue

September is a partial month (data ends 8 Sep) and is not comparable.

Interpretation: Net revenue rose steadily from 1.63M in January to 1.84M in May (+12.7%), dipped to 1.77M in June and 1.54M in July (the lowest full month, 12% below the January-to-August average of about 1.76M), then jumped to 2.03M in August (the peak, +31% on July and about 15% above average). Peak months by order count are August (654) and April (638), and by revenue August and May.

Business implication: Revenue is more volatile than order volume. July had a near-average 595 orders yet the lowest revenue, so its dip came from lower order values or more cancellations, not fewer customers, which is worth investigating. August is the business's strongest month, and the July-to-August swing is large enough to affect stock and capacity planning.

Multivariate analysis



Chart 22: Correlation matrix of order measures

Interpretation: Only revenue-related measures are meaningfully correlated. Order revenue is strongly linked to units ordered (0.89) and moderately to the number of item lines (0.79) and distinct products (0.79), so about 79% of the variation in order revenue is explained by the number of units alone. Every other pair is close to zero: revenue vs delay days (-0.03), revenue vs delayed flag (-0.03), and same-city delivery vs month (-0.03).

Business implication: Order value is driven by basket size. Delivery delay, returns and shipping distance are unrelated to revenue and to each other. This is useful for later modelling, since basket measures are good predictors of order value, while delay and returns will be difficult to predict from the variables available in this data.

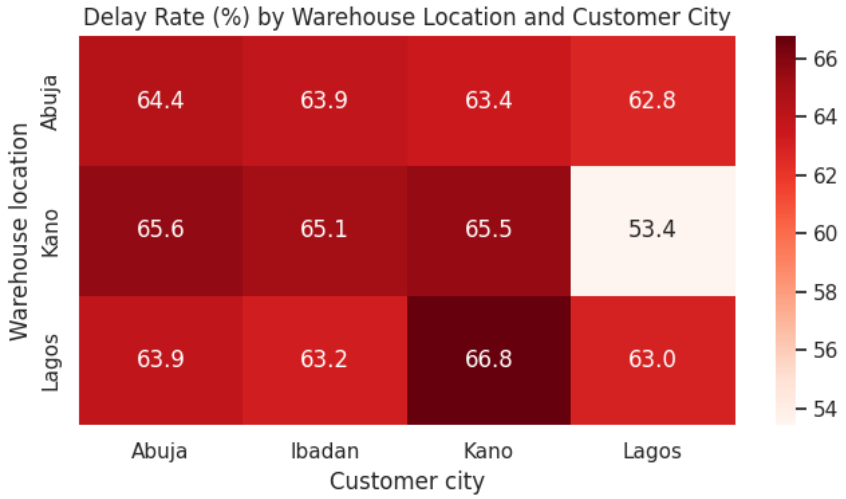


Chart 23: Delay rate by warehouse location and customer city

Interpretation: Most routes sit between 62.8% and 66.8%. The standout is Kano warehouses serving Lagos customers at 53.4%, about 10 points below the company average of 63.7%. This route has 249 orders (about 133 delayed against about 159 expected), so it is not a small sample, and the gap is large enough that chance is an unlikely explanation even after allowing for 12 routes being compared. The highest rates are Lagos warehouses to Kano customers (66.8%), Kano to Abuja (65.6%) and Kano to Kano (65.5%).

Order volumes by route:

Warehouse location	Abuja	Ibadan	Kano	Lagos	Total
Abuja	320	335	328	352	1,335
Kano	250	261	258	249	1,018
Lagos	635	644	701	667	2,647

Business implication: Route does not systematically drive delay: same-city routes are not faster (consistent with section 2.4). The Kano-to-Lagos route is the exception and the only one worth investigating, since whatever is working there (carrier, dispatch process, timing) could be copied to other routes. Note the route has no obvious logistical advantage, which makes the cause more likely to be operational than geographic.

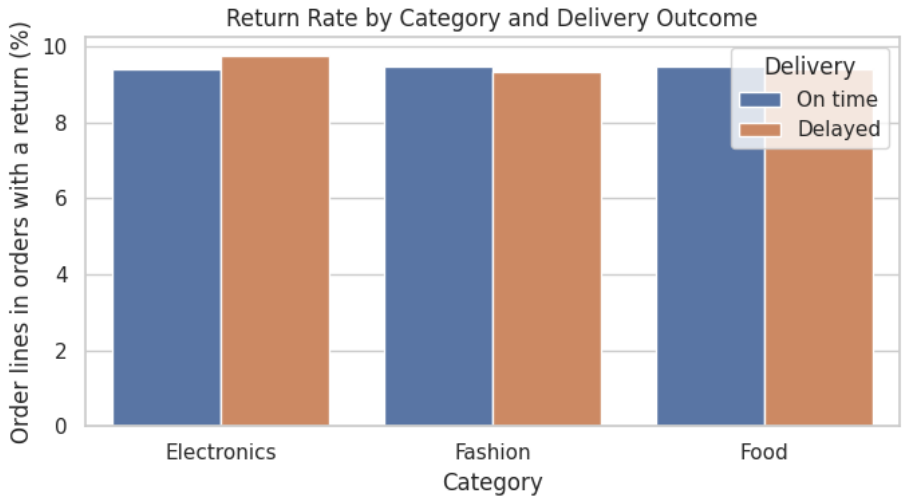


Chart 24: Return rate by category and delivery outcome

Interpretation: Return rates are almost identical across categories and delivery outcomes, between 9.3% and 9.8%. The largest gap between delayed and on-time orders within a category is 0.35 points (Electronics: 9.76% vs 9.41%).

Business implication: No category stands out for returns, and delay does not change that. Returns look like a fairly constant background rate of about 9.5% across the business, so returns reduction should focus on the

stated reasons (late delivery, wrong item, damaged) and on picking and packing quality, not on specific categories. Product-level return rates are unreliable here, because about 500 returns spread across 250 products gives roughly 2 returns per product.

Follow-up: do return reasons match delivery outcomes?

Return reason	Returns	From delayed shipments	From on-time shipments	% from delayed
Damaged	152	88	64	57.9%
Late Delivery	185	119	66	64.3%
Wrong Item	163	110	53	67.5%
All returns	500	317	183	63.4%

Interpretation: Returns are spread across delayed and on-time shipments in the same proportion as shipments overall (63.4% of returns vs 63.7% of shipments are delayed). Orders with a "Late Delivery" return were delayed 64.3% of the time, no more than other return reasons, and 66 of the 185 (35.7%) were classed as On Time (delay of 0 to 2 days). A rough chi-square test finds no significant difference between reasons (p of about 0.2), to be confirmed in Part 4.

Business implication: The recorded return reason does not line up with what happened in delivery, so "Late Delivery" returns cannot be used as evidence of delays caused by the delivery process. There are three possible explanations: customers judge lateness against a promised date that is not in the data, customers see even 1 to 2 days as late (which would mean the 3-day "On Time" tolerance is too generous), or reasons are recorded loosely. In the meantime the return reason field should be treated as unreliable, and a delivery-date promise should be captured so lateness can be measured the way customers experience it.

Summary of Bivariate and Multivariate Findings

1. **Revenue is driven by basket size** (units ordered, r = 0.89) and is spread across customers: the top 10% generate 23.9% of net revenue.
2. **Fashion earns the most in total (37.5%)** because of its larger catalogue, while **Food earns the most per product** (92,382 vs about 84,000 for the others).
3. **Peak activity is in August** (654 orders, 2.03M net revenue), with a July dip in revenue (1.54M) despite near-average order volume.
4. **Delays are company-wide:** warehouse rates range from 58.7% to 68.0% around a 63.7% average, a spread consistent with chance.
5. **Delay is unrelated to distance, workload or returns:** same-city vs cross-city (63.9% vs 63.7%), shipments per employee (r = -0.13), and returns (9.4% vs 9.5%).
6. **Returns run at about 9.5% of orders** across categories and delivery outcomes, with no group standing out.
7. **Stock is not aligned to demand:** W12 and W01 ship about the same volume but hold 77.5k and 56.5k units.
8. **Kano-to-Lagos delivery (53.4% delayed, 249 orders) is the only route that stands out** and is unlikely to be chance. It accounts for nearly all of Lagos's lower delay rate and is tested formally in Part 4.
9. **Supplier vs delivery performance cannot be analysed,** because suppliers link to no other table.
10. **Return reasons do not match delivery outcomes:** 36% of "Late Delivery" returns came from shipments classed as On Time, and returns are spread across delayed and on-time shipments in the same proportion as shipments overall.

5. Statistical Analysis

Descriptive Statistics

Variable	Mean	Median	Mode	Std dev	Variance	Skewness	Kurtosis
Order revenue (orders with items)	4,548	3,949	366.5	3,149	9,915,637	0.99	0.98
Units ordered per order	16.49	15	10	10.77	116.1	0.77	0.63
Delay days	3.57	4	5	2.29	5.25	-0.05	-1.24
Quantity per order line	5.50	6	9	2.89	8.33	-0.01	-1.24
Line revenue	1,445	1,152	261.4	1,151	1,324,769	0.91	0.01
Unit price	264.2	263.6	11.78	140.0	19,601	-0.03	-1.20
Stock per warehouse-product pair	410.5	364	416	283.7	80,467	1.19	1.99

Kurtosis is excess kurtosis (a normal distribution = 0). For continuous variables (order revenue, line revenue, unit price) the mode is just the single most repeated exact value and carries little meaning.

Interpretation

1. **Money and stock are right-skewed.** Order revenue (skew 0.99), line revenue (0.91) and stock (1.19) have means above medians (4,548 vs 3,949 for orders). A minority of large orders and large stock holdings pull the average up, so the **median is the fairer "typical" value** for these variables. Stock also has the heaviest tails (kurtosis 1.99), meaning more extreme values than a normal distribution, which matches the 87 high-stock outliers found earlier.
2. **Delay days, quantity and unit price are flat, not bell-shaped.** All three have skewness near zero and kurtosis near -1.2, which is the signature of a **uniform distribution**: every value from 0 to 7 days, 1 to 10 units, or 12 to 497 in price is about equally likely. Real operations rarely look like this, which supports the earlier observation that the data is probably simulated.
3. **Variability is high relative to the mean for revenue** (standard deviation 3,149 vs mean 4,548, about 69%), so order values are quite unpredictable. Delay is similarly spread: a standard deviation of 2.29 days around a 3.57-day mean.

Business implication: Use medians when reporting typical order value or stock. Because delay is spread evenly across 0 to 7 days, there is no "usual" delay to aim at, and averages hide the full spread. For future modelling, revenue and stock would benefit from a log transform to reduce skew.

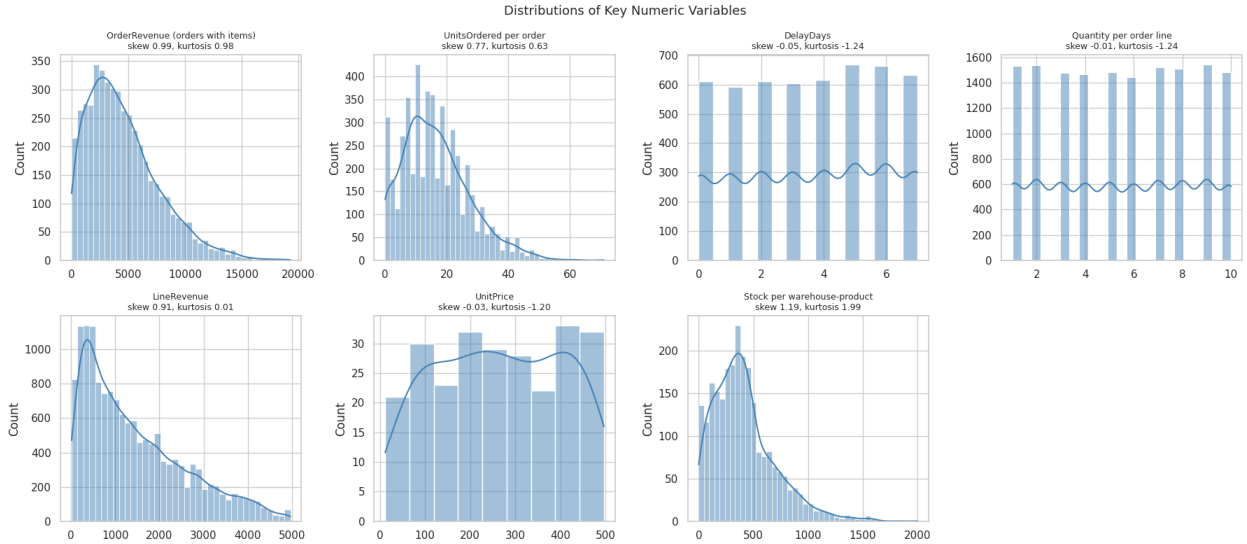


Chart 25: Distributions of key numeric variables, with skewness and kurtosis

Distribution and Correlation Analysis

Chart 25 (distributions grid, saved as `25_distributions_grid.png`) shows each variable's histogram with its skewness and kurtosis.

Interpretation: Revenue, units and stock have a long right tail, while delay days, quantity and unit price are flat across their range. No variable is approximately normal.

Business implication: Because the key operational measures are not normally distributed, tests that assume normality (such as ANOVA) were avoided. A Kruskal-Wallis test was used for delay days, and chi-square tests for proportions.

Correlation at order-line level

	Quantity	Unit price	Line revenue
Quantity	1.00	0.005	0.66
Unit price	0.005	1.00	0.67
Line revenue	0.66	0.67	1.00

Interpretation: Quantity and unit price are unrelated to each other (0.005), and each drives line revenue about equally (0.66 and 0.67). That is expected, since line revenue is quantity multiplied by price.

Business implication: Customers do not buy more of cheaper items, so volume and price contribute independently. Revenue grows by raising the units per order or the value of items sold, and each lever is worth about the same. This is consistent with the earlier finding that revenue at order level is driven by basket size (units ordered r = 0.89), because larger baskets add more lines.

Hypothesis Tests

All tests use a significance level of 0.05. Chi-square tests are used for rates (proportions), and Kruskal-Wallis for delay days because they are not normally distributed. Cramér's V measures effect size (0 = no relationship, 1 = perfect; below 0.1 is negligible).

#	Question	Test	Statistic	p-value	Cramér's V	Decision
1	Does the delay rate differ across the 15 warehouses?	Chi-square	15.76 (df 14)	0.328	0.056	No significant difference
2	Do delay days differ across warehouses?	Kruskal-Wallis	H = 10.04	0.759	n/a	No significant difference
3	Does the delay rate differ across customer cities?	Chi-square	6.27 (df 3)	0.099	0.035	No significant difference
4	Do delayed orders have a different return rate?	Chi-square	0.01 (df 1)	0.928	0.001	No significant difference
5	Is return reason related to delivery outcome?	Chi-square	3.23 (df 2)	0.199	0.080	No significant difference
6	Does any single route differ from the rest?	Chi-square per route, Bonferroni-corrected (12 comparisons)	see below	see below	n/a	**Kano to Lagos is significant**

Test 1: Delay rate across warehouses

H0: the delay rate is the same in all warehouses. **H1:** at least one warehouse differs. **Result:** p = 0.328, so H0 is not rejected. The apparent range of 58.7% (W01) to 68.0% (W05) is within what chance produces with about 330 shipments per warehouse. **Business meaning:** No warehouse is a proven under-performer or over-performer on delay rate. Delay is a company-wide issue, so fixes should target shared processes, not individual warehouses.

Test 2: Delay days across warehouses

H0: delay days are distributed the same in every warehouse. **H1:** at least one differs. **Result:** H = 10.04, p = 0.759, so H0 is not rejected. **Business meaning:** Delay severity (not just the delay rate) is also the same everywhere. Test 1 and Test 2 agree, which strengthens the conclusion that warehouses perform alike.

Test 3: Delay rate across customer cities

H0: the delay rate is the same in every customer city. **H1:** at least one city differs. **Result:** p = 0.099, so H0 is not rejected, though the result is close to the 0.05 line. The effect is negligible (V = 0.035). **Business meaning:** Delivery to Kano (65.7%) and Lagos (61.0%) do not differ reliably. As the route test shows, Lagos's lower rate is driven by a single route and not by the city overall.

Test 4: Return rate, delayed vs on-time orders

H0: the return rate is the same for delayed and on-time orders. **H1:** it differs. **Result:** p = 0.928, which is effectively no difference (9.42% vs 9.54%). **Business meaning:** Reducing delays should not be expected to reduce returns. Returns need to be addressed through order accuracy and handling.

Test 5: Return reason vs delivery outcome

H0: return reason is independent of whether the shipment was delayed. **H1:** they are related. **Result:** p = 0.199, so H0 is not rejected. "Late Delivery" returns (64.3% from delayed shipments) are no more likely to come from delayed shipments than "Wrong Item" (67.5%) or "Damaged" (57.9%) returns. **Business meaning:** The recorded return reason does not match what happened in delivery, so return reasons are unreliable for diagnosing delivery problems. Recording the promised delivery date and validating the reason field would fix this.

Test 6: Delay rate by route (12 routes, Bonferroni-corrected)

H0: a route's delay rate equals the delay rate of all other routes. **H1:** it differs. Testing 12 routes at once raises the chance of a false positive, so p-values were multiplied by 12 (Bonferroni correction).

Route	Orders	Delay rate	Raw p	Adjusted p	Significant
Kano to Lagos	249	**53.4%**	0.0007	**0.008**	**Yes**
Lagos to Kano	701	66.8%	0.078	0.933	No
All other 10 routes	250 to 667	62.8% to 65.6%	0.57 to 0.996	1.000	No

Result: Only **Kano to Lagos** differs from the rest, and the difference survives the correction (adjusted p = 0.008). Its delay rate is about 10 points below the company average of 63.7%. **Business meaning:** This is the one route doing something different, and the only positive finding in the delivery data. Whatever is working there (carrier, dispatch process, loading) should be identified and tried on other routes. It should be treated as a lead to investigate and not a proven cause, since the data holds no operational detail to explain it.

Summary of Statistical Findings

1. **Five of six tests found no significant difference.** Delay rate and delay days do not differ across warehouses or (significantly) across cities, and returns do not differ by delay outcome.
2. **Delay is a system-wide problem.** With about 64% of shipments delayed everywhere, improvement should come from shared processes and not from individual warehouses.
3. **Kano to Lagos is the one exceptional route** (53.4% delayed, adjusted p = 0.008) and the best lead for improvement.
4. **Return reasons do not reflect delivery outcomes** and should be validated.
5. **Revenue and stock are right-skewed**, so medians are used for "typical" values. Delay, quantity and price are uniformly spread, which suggests the data may be simulated.

6. Feature Engineering

The feature table below explains why each feature was created, its business value and how it could improve machine learning models. The final dataset, `Feature_Engineered_Dataset.csv`, holds one row per order.

Feature catalogue

Feature	How it was built	Why created	Business value	How it helps ML
DeliveryDuration	Copy of `DelayDays` (no ship or delivery dates exist, so delay days is the only time measure)	Measure how late each delivery was	Shows how long customers wait beyond schedule	Regression target for delay length; input for return or satisfaction models
IsDelayed (delay flag)	1 if `DelayDays` >= 3 (inferred rule, matches supplied Status on all 5,000 rows)	Turn delay into a yes/no outcome	Gives a single on-time KPI	Classification target for delay prediction
AvgCustomerSpend	Net revenue ÷ net orders per customer	Capture typical basket value	Identifies high-value buyers	Predictor for customer value and churn models
OrderFrequency	Number of orders per customer	Capture buying habit	Separates loyal from occasional buyers	Strong predictor of future orders and CLV
CLV_Estimate	Net revenue ÷ months observed x 12 (annualised; ignores margin and churn)	Estimate what a customer is worth per year	Prioritises retention spending	Regression target or input for customer-value models
ReturnFrequency	Orders with a return ÷ orders per customer	Capture return behaviour	Flags customers or patterns costing money	Predictor in return-prediction models
InventoryTurnover	Units sold ÷ stock held, per warehouse	Show how fast stock moves	Highlights overstocked or slow warehouses	Input for demand forecasting and replenishment
WarehouseUtilisationIndex	Stock held ÷ highest warehouse stock x 100 (proxy, since capacity was not provided)	Estimate how full each warehouse is relative to the largest	Guides stock rebalancing	Input for warehouse performance models
WarehouseOnTimeRate	On-time share of a warehouse's shipments (stand-in for supplier reliability, which cannot be built because suppliers link to no table)	Measure dependability	Ranks warehouses for review	Input for reliability and delay models
DeliveryEfficiencyScore	Half on-time rate, half (1 − average delay ÷ 7), scaled 0 to 100	Combine frequency and severity of delay	One comparable score per warehouse	Compact input; target for warehouse performance
SeasonalDemandIndex / PeakMonthFlag	Orders per day in a month ÷ average orders per day (September adjusted for 8 days); peak if >= 1.05	Show which months run hot	Supports staffing and stock planning	Captures seasonality in demand forecasting
DemandTrendPct / DemandTrend3M	Month-on-month change and 3-month rolling average of orders per day	Show direction of demand	Early warning of growth or decline	Trend input for forecasting

Caution for modelling: customer and warehouse features are calculated over all orders, including the order being predicted, which leaks information into a model. Before training, they should be rebuilt using only data from before each order. Features also rest on one 8-month window with an incomplete inventory table, so they are indicative.

Final dataset: `Data/Feature_Engineered_Dataset.csv`, one row per order, with the original merged fields plus the engineered features above.

Feature Engineering Results

Final dataset: `Data/Feature_Engineered_Dataset.csv`, 5,000 rows (one per order) x 43 columns.

The 1,765 missing values are expected. They come only from the trend features: `DemandTrendPct` is blank for January (594 orders, no prior month) and `DemandTrend3M` is blank for January and February (594 + 577

orders, a 3-month window needs three months). 594 + 1,171 = 1,765. They were left blank on purpose, since filling them would invent data. Models can drop those two columns or those months.

Order-level summary of key features (customer features repeat on each of a customer's orders, so frequent buyers weigh more: mean 6.10 orders vs 5.05 at customer level)

Feature	Mean	Range
OrderFrequency	6.10	1 to 15
AvgCustomerSpend	4,246	0 to 13,077
CLV_Estimate (annualised)	25,733	0 to 85,710
ReturnFrequency	0.09	0 to 1.00
DeliveryEfficiencyScore	42.6	39.0 to 45.9
WarehouseUtilisationIndex	86.7	72.9 to 100
InventoryTurnover	0.04	0.03 to 0.04
SeasonalDemandIndex	1.00	0.96 to 1.07

Warehouse findings

- **Delivery efficiency scores are tightly bunched** (39.0 to 45.9). The best are W01 (45.9), W10 (45.2) and W07 (45.0), and the weakest are W05 (39.0), W09 (40.0) and W03 (40.0). This agrees with Test 1: warehouse differences are small and not statistically significant.
- **Utilisation proxy** (stock held vs the largest warehouse): W12 is 100 (77,517 units) and W01 is lowest at 72.9 (56,539 units). This is an estimate, since capacity was not provided.
- **Inventory turnover is very low.** The warehouses hold about 1.0 million units against about 36,400 units sold in 8 months (net orders), roughly **28 times** what was sold, a turnover of about 0.04. This is understated, because only shipped pairs that have an inventory record are counted (about two thirds of them). Even adjusting for that, turnover would be about 0.055. Either SwiftLog is heavily overstocked, or the inventory table does not reflect true on-hand stock. Treat it as indicative.
- **Workload:** W14 (37.8), W15 (35.2) and W12 (33.7) handle the most shipments per employee, against about 20 to 25 at most others.

Demand features (orders per day, so February's 28 days and September's 8 days are comparable)

Month	Orders per day	Seasonal index	Peak flag
Apr	21.27	1.07	Yes
Aug	21.10	1.06	Yes
Feb	20.61	1.04	No
All other months	19.2 to 19.7	0.96 to 0.99	No

Seasonality is weak (every month is within about 7% of average) and the 3-month trend is flat (19.4 to 20.4 orders per day). Per day, February is among the stronger months, which refines chart 03, where it looked lowest on raw order count only because it has fewer days.

Business implication: Demand planning can assume stable volume, with modest peaks in April and August. The inventory features point to a stock problem (overstock or unreliable records) more than a delivery problem. The delivery-related features show little spread between warehouses, which will limit how well delay can be predicted from them.

7. Segmentation and Pattern Discovery

Customer segmentation (K-Means)

Method: K-Means on four standardised features (order frequency, average spend, net revenue, return frequency) for the 991 customers who ordered. Silhouette scores were 0.284 (k=2), 0.308 (k=3), 0.308 (k=4) and 0.287 (k=5). k=3 was chosen (tied with 4, but simpler to act on). Segments are ranked by average net revenue, S1 highest. Charts: `26_customer_segments_pca.png` and `27_segment_customer_vs_revenue_share.png`.

Segment	Customers	% of customers	Avg orders	Avg spend per order	Net revenue per customer	% of net revenue	Return freq.	Days since last order
S1: Loyal high-value	351	35.4%	7.06	5,277	24,350	**59.2%**	0.07	34
S2: Return-prone mid-value	215	21.7%	4.64	3,743	11,875	17.7%	**0.31**	48
S3: Low-frequency, low-value	425	42.9%	3.59	3,415	7,857	23.1%	0.00	**61**

Interpretation:

1. **S1 is the core of the business:** 35% of customers produce 59% of net revenue, about 3.1x the revenue of an S3 customer, buying twice as often at higher order values, with few returns, and ordering most recently.
2. **S2 buys moderately but returns heavily:** about 0.31 returns per order, more than four times S1's rate.
3. **S3 is the largest group and the weakest:** it has the fewest orders, lowest value, no returns, and has gone longest since its last order (61 days), so it is the most likely to be lapsing.
4. **Caution:** the silhouette score of 0.31 shows moderate overlap, so use the segments for targeting and not as hard boundaries. S2 and S3 are separated mainly by return behaviour, which the earlier tests found to be close to random in this data, so those two should not be over-interpreted.

Business use:

- **S1:** priority service and loyalty rewards, and protect these customers from delays and cancellations.
- **S2:** investigate order accuracy and product quality on their orders, and follow up after returns.
- **S3:** win-back campaigns and bundles to increase basket size (units per order is the main driver of order value, r = 0.89).

Warehouse segmentation (K-Means, k = 3)

Inputs: delay rate, shipments per employee, total stock, inventory turnover. Chart: `28_warehouse_segments.png`. With only 15 warehouses this is exploratory.

Segment	Warehouses	Delay rate	Shipments per employee	Avg stock	Turnover
Lowest delay	W01, W07	59.6%	22.0	62,439	0.041
Typical (10 warehouses)	W02, W03, W04, W05, W06, W08, W09, W10, W11, W13	64.8%	23.8	67,396	0.034
High workload per employee	W12, W14, W15	63.1%	35.6	69,641	0.039

Interpretation:

1. **W01 and W07** have the lowest delay rate and the best turnover. They were grouped after the fact, and Test 1 found warehouse differences overall are not significant, so they are a watch-list for good practice and not a proven result.
2. **W12, W14 and W15** carry about 50% more shipments per employee than the typical group (35.6 vs 23.8) with a similar delay rate (63.1% vs 64.8%).

Business implication: Heavier workload per employee does not show up as more delay, so the typical group may be over-staffed relative to W12, W14 and W15, or those three are running efficiently. A monitored staffing pilot would settle this. Supplier and regional segmentation were not possible: suppliers link to no other table,

and regional differences are not significant.

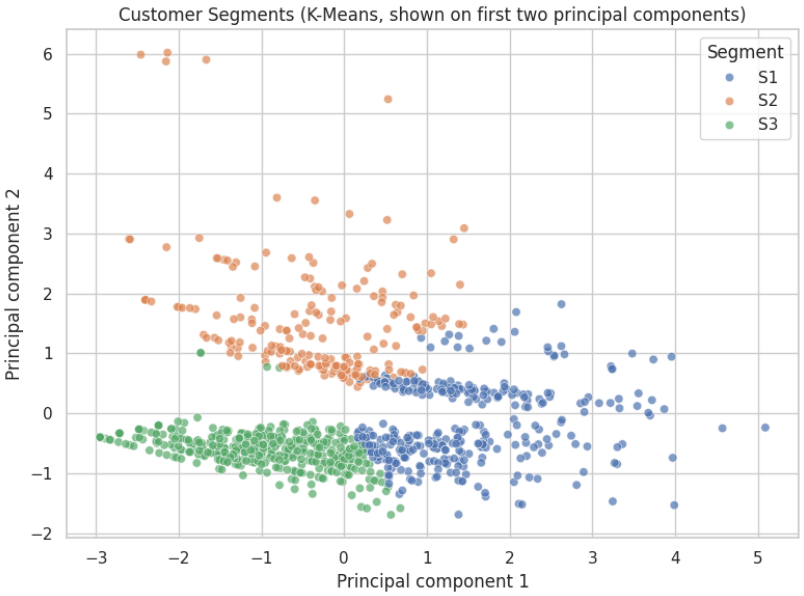


Chart 26: Customer segments (K-Means) on the first two principal components

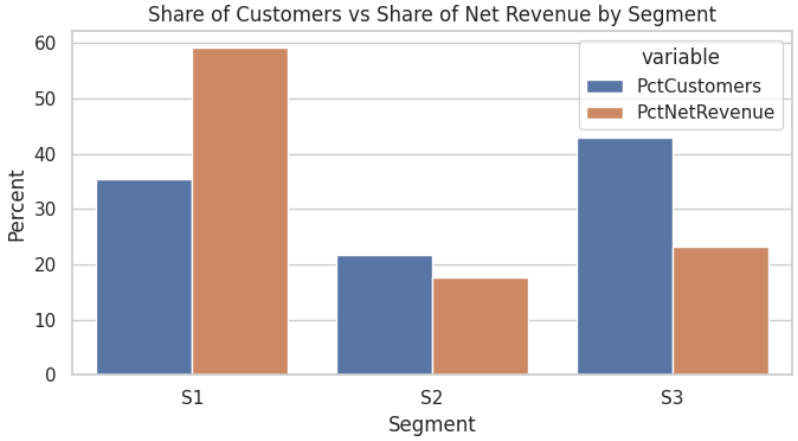


Chart 27: Share of customers vs share of net revenue by segment

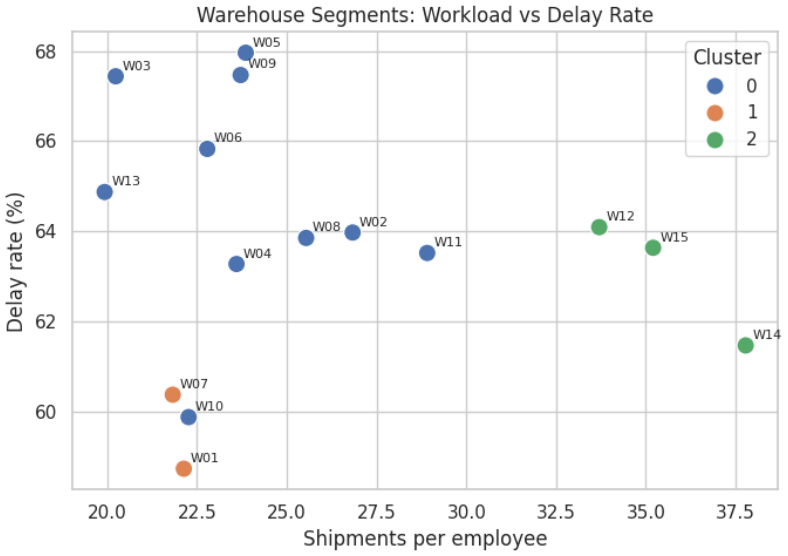


Chart 28: Warehouse segments, workload vs delay rate

8. Business Insights

Answers to the ten business questions

#	Question	Answer and evidence
1	Which customer segments generate the highest value?	**S1, Loyal high-value:** 351 customers (35.4%) produce 59.2% of net revenue, average 7.1 orders, 24,350 net revenue per customer, low returns (0.07) and the most recent activity (34 days).
2	Which products contribute most to revenue?	By category, **Fashion leads** (5.41M net, 37.5%), then Food (32.8%) and Electronics (29.7%). Fashion leads because it has the most products (95), while **Food earns the most per product** (92,382 vs 84,838 for Fashion and 83,270 for Electronics). The top-10 product table above lists individual products.
3	Which suppliers consistently underperform?	**Cannot be determined.** The suppliers table (150 records, ID and name only) has no link to any other table, so no supplier performance exists in the data. See Assumptions and Data Limitations.
4	Which warehouses need operational improvement?	**W05** (68.0% delayed, longest average delay 3.79 days, lowest efficiency score 39.0), **W09** (67.5%) and **W03** (67.4%) are the weakest on delay, but differences between warehouses are not statistically significant ($p = 0.328$), so these are watch-list items. **W01** is the thinnest on stock (56,539 units, lowest utilisation proxy), and **W14, W15 and W12** carry the heaviest workload per employee.
5	Which products have the highest return rates?	Returns are about **9.5% of orders** and almost identical across categories (9.3% to 9.8%). With about 500 returns across 250 products (about 2 each), and returns tied to orders and not products, product-level rates are unreliable. The return-incidence table above is indicative only.
6	Which regions see the most delayed deliveries?	Kano customers (65.7%) and Abuja (64.4%) are highest and Lagos (61.0%) lowest, but the city difference is not significant ($p = 0.099$). The one significant finding is a **positive** one: **Kano warehouses to Lagos customers** at 53.4% (adjusted $p = 0.008$). Lagos to Kano is highest at 66.8% but not significant.
7	Which operational bottlenecks were found?	(a) **63.7% of shipments are 3+ days late**, spread evenly across all warehouses; (b) only **33% of orders are delivered**, 32.5% are cancelled and 34% are pending; (c) cancelled orders hold **7.2M, about a third of gross revenue**; (d) **34.4% of shipped warehouse-product pairs have no inventory record**; (e) stock is about 28 times units sold, and 10.8% of stocked pairs are below 100 units; (f) return reasons do not match delivery outcomes.
8	Which variables suit predictive modelling?	**Order revenue** is predictable from units ordered ($r = 0.89$) and item lines (0.79). **IsDelayed** is a balanced target (63.7% vs 36.3%), but no variable in the data correlates with it (all $\lvert r \rvert < 0.05$), apart from the Kano-to-Lagos route. **HasReturn** is imbalanced (9.5%) and also unrelated to delay. Customer segment, order month, route and warehouse are the best candidates, with realistic expectations of modest accuracy for delay and return models.
9	What hidden patterns were found?	(a) Kano-to-Lagos is the only route that differs (about 10 points better); (b) delay is evenly spread over 0 to 7 days, with no link to distance, workload or returns; (c) 36% of "Late Delivery" returns came from shipments classed as On Time; (d) customer value is driven by order frequency and basket size, and the lowest-value segment is also the most dormant; (e) stock levels are unrelated to what each warehouse ships; (f) July revenue dipped 12% below average despite normal order volume, then August peaked.
10	What recommendations would improve operations?	See section 7.2.

Supporting product tables

Top products by net revenue. The ten best sellers account for only 8.4% of net revenue, so revenue is not dependent on a few products.

Product	Category	Net revenue	Units
P085	Food	143,028	290
P040	Electronics	131,276	277
P025	Food	123,456	266
P159	Fashion	120,645	252
P186	Fashion	119,982	260
P229	Food	119,882	242
P209	Fashion	116,844	244
P220	Food	114,378	252
P156	Food	113,149	272
P026	Fashion	112,535	267

Products with the highest return incidence (share of order lines that sit in a returned order). This is **indicative only**: returns carry no ProductID, so a return is attributed to every product in the returned order, and each product has only 47 to 73 lines.

Product	Category	Order lines	Lines in returned orders	Incidence %
P032	Fashion	47	10	21.3
P017	Fashion	53	10	18.9
P108	Electronics	60	11	18.3
P020	Food	59	10	16.9
P177	Electronics	66	11	16.7
P040	Electronics	73	12	16.4
P231	Food	67	11	16.4
P181	Food	61	10	16.4
P109	Electronics	68	11	16.2
P013	Electronics	56	9	16.1

9. Recommendations

Thirteen recommendations follow. Each is tied to a finding in this report. The first three address the largest and best-evidenced problems.

Strategic recommendations

#	Recommendation	Area	Evidence from the analysis
1	**Study and replicate the Kano-to-Lagos route process** (carrier, dispatch timing, handling) and trial it on other routes	Delivery optimisation	53.4% delayed vs 63.7% average; 249 orders; adjusted p = 0.008 (Test 6)
2	**Launch a company-wide delivery improvement programme** with a delay target, not warehouse-by-warehouse fixes	Logistics optimisation	63.7% delayed; no significant differences across warehouses (p = 0.328) or cities (p = 0.099)
3	**Record promised delivery dates and ship/delivery timestamps**, and revisit the 3-day "on time" tolerance	Performance monitoring	Only `DelayDays` exists; 36% of "Late Delivery" returns came from On-Time shipments
4	**Investigate and reduce cancellations**, capturing a cancellation reason	Operational cost reduction	32.5% of orders cancelled, about 7.2M (a third of gross revenue)
5	**Set an ageing rule for pending orders** and review the backlog	Operational efficiency	34.3% of orders pending across an 8-month period
6	**Reconcile inventory records with shipments** and set reorder levels per product	Inventory management	34.4% of shipped pairs have no inventory record; 10.8% of stocked pairs under 100 units
7	**Right-size stock** and rebalance between warehouses, starting with the highest-stock sites	Inventory management	Stock about 28 times units sold (indicative); W12 holds 77.5k vs W01 56.5k units with near-equal shipments
8	**Pilot a monitored staffing rebalance** between the typical group and W12/W14/W15	Warehouse efficiency	35.6 vs 23.8 shipments per employee with similar delay rates (63.1% vs 64.8%)
9	**Protect and reward S1 customers; run win-back and basket-building offers for S3**	Customer retention	S1 = 35% of customers and 59% of net revenue; S3 = 43% of customers, 61 days since last order
10	**Reduce returns through picking verification and packaging checks, and follow up with S2 customers**	Return reduction	Wrong Item (32.6%) and Damaged (30.4%) are 63% of returns; S2 returns 0.31 per order
11	**Build a weekly KPI report:** on-time rate, cancellation rate, pending age, return rate, stock turnover	Performance monitoring	These are the measures on which the problems above were found
12	**Expand and promote the Food range; plan capacity for April and August peaks**	Business growth	Food earns the most per product (92,382); April and August have the highest daily demand (index 1.07 and 1.06)
13	**Close the data gaps:** link suppliers to products or shipments, capture warehouse capacity, validate return reasons	Data quality	Supplier performance, utilisation and return causes could not be measured

10. Conclusion

SwiftLog's data is structurally sound but shows a business with a **system-wide delivery problem** (about two in three shipments 3+ days late), **high cancellation and pending rates**, and **inventory records that do not reconcile with shipping**. Delay is not explained by distance, workload, warehouse or returns, so the cause lies in processes the data does not record. The strongest leads are the Kano-to-Lagos route, which performs significantly better, and the S1 customers, who produce 59% of net revenue. Many variables are evenly distributed, which suggests the data may be simulated, so non-significant results should be read with caution. The engineered dataset is ready for modelling, with the caveat that its customer and warehouse features must be rebuilt without leakage before training.

Future machine learning opportunities

The engineered dataset is a foundation for modelling, with realistic expectations set by what the data shows.

Use case	Target	Best predictors found	Realistic expectation
Order value prediction	OrderRevenue	UnitsOrdered (r = 0.89), item lines and distinct products (0.79)	Strong; basket size explains about 79% of variation
Delivery delay classification	IsDelayed (63.7% vs 36.3%)	Route (Kano to Lagos only); no other variable has abs(r) above 0.05	Low to modest until promised dates and timestamps are captured
Return prediction	HasReturn (9.5% positive)	Customer return history (S2); nothing else	Low; imbalanced target needs class weighting; product-level returns needed
Customer value	Future-period net revenue	OrderFrequency, AvgCustomerSpend, segment	Moderate to high, using a future target to avoid circularity with CLV_Estimate
Demand forecasting	Orders per day by month	Seasonality is weak (index 0.96 to 1.07)	Low until at least 24 months of history exist

Appendix A. Assumptions and Data Limitations

SwiftLog's data is clean in structure (no missing values, valid keys, zero orphan records) but incomplete in content. This section lists what could not be measured, the assumptions made in its place, and the programme guidance followed.

A. Data limitations

#	Limitation	Impact	How it was handled
1	**Suppliers table is isolated.** 150 records with only an ID and a name, and no table contains a `SupplierID`.	Supplier performance, supplier vs delivery analysis and supplier segmentation are not possible.	Treated as a standalone reference table (confirmed acceptable by the programme). Warehouse on-time rate is used as a stand-in reliability feature.
2	**No warehouse capacity data.**	True utilisation cannot be calculated.	Proxy `WarehouseUtilisationIndex` = stock held ÷ largest warehouse's stock x 100. It is an estimate.
3	**The supplied data dictionary is a file list only.**	Field meanings are not documented.	Meanings inferred from column names and the data.
4	**Only `OrderDate` exists.** No ship date, delivery date or promised date.	Delivery time and lateness against customer expectations cannot be measured.	`DelayDays` is the only time measure, and `DeliveryDuration` is a copy of it.
5	**Returns have no `ProductID` and no date.**	Product-level return rates are unreliable (about 500 returns across 250 products, roughly 2 each).	Returns are analysed per order (`HasReturn`). Category return rates are measured at order-line level, so a return counts against every product in the returned order.
6	**Period is 1 Jan to 8 Sep 2026 only.**	No year-on-year comparison, and September is a partial month.	Seasonality uses orders per day, September is excluded from month comparisons, and seasonal conclusions are limited.
7	**Inventory has no dates and no in/out direction.**	It is unclear whether movements add or remove stock, so inventory turnover is indicative only.	Movements are summed per warehouse-product pair (programme guidance).
8	**Inventory does not reconcile with shipments.** 34.4% of shipped warehouse-product pairs have no inventory record, matching the 34.5% expected if shipping ignored inventory.	Stock-based metrics (turnover, stock-outs) may not reflect true on-hand stock.	Stock-based figures are reported as indicative.
9	**Order status contradicts other tables.** Cancelled orders have shipments, and returns come from Pending and Cancelled orders in equal proportion.	Status cannot be used to filter other tables.	Each table is used as recorded. Revenue follows order status, and delivery follows shipment records.
10	**No currency stated.**	Amounts have no unit.	Assumed to be Naira, since SwiftLog is a Nigerian company.
11	**The data appears simulated.** Delay days, quantity and unit price are uniformly distributed, and most categories are almost equal in size.	Non-significant results may reflect how the data was built and not how the business runs.	Findings are stated with caution, and null results are reported as such.

B. Assumptions

Definitions and calculations

#	Assumption	Basis
1	**Delayed means 3 or more days late** (`DelayDays >= 3`).	Inferred from the data: matches the supplied delivery Status on all 5,000 shipments (On Time = 0 to 2 days, Delayed = 3 to 7).

#	Assumption	Basis
2	**Line revenue = Quantity x UnitPrice**, and order revenue is the sum of its lines. **Net revenue** (excluding Cancelled orders) is the headline figure, with Gross (all orders) and Delivered-only also reported.	No revenue column exists.
3	**The 236 orders with no items** have zero revenue, carry a `HasItems = 0` flag, and are excluded from average order value but kept in order counts.	They have no rows in `order_items`.
4	**Return rate = orders with at least one return ÷ all orders** (about 9.5%). All 500 return rows are kept, including 27 orders with two returns (5 repeating the same reason).	Each return has a unique `ReturnID` and there are no dates to prove a repeat is a duplicate.
5	**Low stock means fewer than 100 units** per warehouse-product pair.	No reorder level was provided.
6	**Workload = shipments per employee**, with each shipment attributed to its dispatching warehouse.	The Employees table links only to warehouses.
7	**"Region" means customer city** (destination). Warehouse location is used where the origin matters.	No separate region field exists.
8	**Seasonal demand index = orders per day ÷ average orders per day.** September is counted as 8 days, and a peak month has an index of at least 1.05.	Months have different lengths, and September is partial.
9	**Customer lifetime value = net revenue ÷ months observed x 12** (annualised).	A simple estimate that ignores margin and churn.
10	**Delivery efficiency score** = 50% on-time rate + 50% (1 − average delay ÷ 7), scaled 0 to 100.	An equal-weight composite chosen for comparability.

Data preparation

#	Assumption	Basis
11	**Repeated warehouse-product inventory rows are stock movements and are summed** (4,000 rows to 2,455), including 3 fully identical rows.	Programme guidance.
12	**High values were kept as outliers:** 119 order lines and 87 stock records outside 1.5 x IQR.	They are legitimate results of price x quantity and of summed stock movements.
13	**Text standardisation changed 0 values,** since labels were already consistent.	Run as a validation step.
14	**Two merged datasets were built:** `order_master` (one row per order) and `item_master` (one row per order line). Items and returns were aggregated before joining.	Avoids repeating order-level values across items.
15	**Encoded data is kept separate** (`order_master_encoded`), `DeliveryStatus` is dropped from it as it duplicates `IsDelayed`, and no scaling was applied.	Keeps the analysis dataset readable and avoids leakage.

Analysis

#	Assumption	Basis
16	**Chart 13 uses net revenue and chart 14 uses gross revenue,** as stated in each chart's note.	Chart 14 compares products on everything customers ordered.
17	**Return reasons are treated as unreliable** for diagnosing delivery problems.	36% of "Late Delivery" returns came from shipments classed as On Time, and reason is not related to delay ($p = 0.199$).
18	**Tests use a 5% significance level.** Chi-square tests for proportions, Kruskal-Wallis for delay days, and a Bonferroni correction for the 12 route comparisons.	Delay days are not normally distributed.
19	**Segmentation is exploratory.** The customer silhouette score is 0.31 (k=3 and k=4 tied, k=3 chosen for simplicity), and the warehouse segmentation uses only 15 observations.	Segments overlap and are for targeting, not hard boundaries.

#	Assumption	Basis
20	**Engineered features were calculated over all orders,** so they include the order being predicted.	They must be rebuilt using only earlier data before any predictive model is trained, to avoid leakage.

C. Programme guidance followed

1. Suppliers are treated as a standalone reference table, and no link is invented.
2. A proxy measure is used for warehouse utilisation and labelled as an estimate.
3. Repeated inventory rows are treated as stock movements, summed and not removed.
4. Definitions missing from the dictionary (such as "Delayed" vs "On Time") are inferred and documented here.

D. Recommended data improvements for SwiftLog

1. Add a `SupplierID` to products or shipments so supplier performance can be measured.
2. Record warehouse capacity so utilisation is measured and not estimated.
3. Capture order, ship, promised and actual delivery dates.
4. Add `ProductID` and a date to returns, and validate the return reason field.
5. Date-stamp inventory transactions and record whether each adds or removes stock.
6. Record a cancellation reason, and reconcile order status with shipment and return records.